Ge'ez Morphological Analyser (ByT5): Training Report, Run A

geez-morphology-model commit a1ad2b6 · geez-dataset commit 04f45a7 · Kaggle notebook esubechekol/geez-byt5-gpu-v2-a · report built 18 October 2026

Result. Run A trained to completion (10,980 steps, 10 epochs); the checkpoint kept is from step 5,000. On the **held-out test set** (2,310 words whose roots never occur in training) it reaches **89.8% root accuracy**, **67.4% exact full analysis** (root and every feature) and **85.7% part-of-speech accuracy** (selection score 0.786, against 0.809 on validation). This model is the label source for the next translation experiments.

1. What the model does

The analyser reads one Ge'ez word, character by character (bytes), and writes its morphological analysis as text. It is trained on three tasks at once, marked by a task prefix: **analyse** (word → root + pattern features), **generate** (root + features → word) and **part of speech** (word → word class). For example, ሰገዱ should be analysed as root s-g-d with perfective, 3rd person, plural, masculine features, and tagged as a verb.

Item	Value
Architecture	ByT5-small encoder–decoder (T5, gated-GELU): 12 encoder / 4 decoder layers, d_model 1472, feed-forward 3584, 6 heads, byte vocabulary 384
Run type	A = fine-tuning from the public ByT5-small checkpoint (no Ge'ez pretraining). Run B adds Ge'ez span-corruption pretraining first.
Hardware	Kaggle, 2 × NVIDIA T4, PyTorch 2.11.0+cu128 DDP, fp32
Batch	32 rows × 2 GPUs × 8 accumulation = 512 rows per step; 1,098 steps per epoch
Schedule	10,980 steps = 10 epochs; warm-up 500 steps; logged learning rate peaks at 1.0e-03 and decays to 1e-04
Model selection	every 1,000 steps on the validation set; score = mean of exact-analysis accuracy and root accuracy; best checkpoint kept in best/
Wall-clock	three Kaggle sessions of about 11 h each (session 1: 11.3 h, steps 1–3,676; session 2: steps 3,677–7,530; session 3: 10.8 h, steps 7,531–10,980 plus the final report), about 33 h in total, at a median 0.092 steps/s

2. Training data

All data come from the author's geez-dataset (commit 04f45a7). Before training, the script checks that the data load exactly as expected and that no validation or test root, and no validation or test word form, appears in training.

Source	Word pairs	What it is
synthetic	118,131	forms generated from checked roots and paradigm rules
unified_book	10,002	forms transcribed from printed grammars and books (unified table)
gold_nominal	6,168	hand-checked noun/adjective paradigm cells
gold_verb	1,727	hand-checked verb paradigm cells
dictionary	1,080	dictionary headwords with checked roots
extra_synthetic	247	additional checked generated cells
dictionary_verb	59	dictionary verbs with checked roots
word_without_root	32	function words labelled as rootless
total	137,446	expanded to 562,403 training samples over the three tasks

Leakage audit (from the run). Training: 562,403 samples, 1,946 distinct roots. Validation: 1,736 pairs, 147 roots; test: 2,310 pairs, 217 roots. Training pairs sharing a root, a word form, or a spelling-variant root with validation or test: **0 in every check** (RESULT: CLEAN).

Class balance. Verbs dominate the training samples: verb 422,691 (75.2%), adjective 68,400 (12.2%), noun 56,196 (10.0%), nominal 9,200 (1.6%); all other classes together 1.1%. Synthetic forms are 86% of word pairs.

3. Training curve

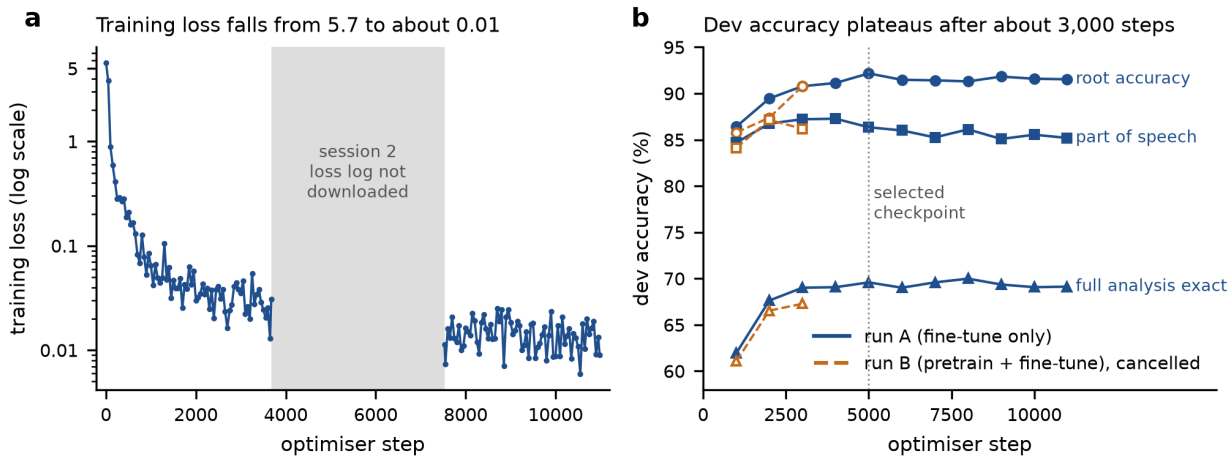


Figure 1. (a) Training loss per logged step; the grey band marks session 2 (steps 3,677–7,530), whose training-loss log was not downloaded. (b) Validation accuracy every 1,000 steps for run A (solid) and the cancelled run B (dashed, steps 1,000–3,000 only). The dotted line marks the selected checkpoint (step 5,000, selection score 0.8088).

Step	1k	2k	3k	4k	5k	6k	7k	8k	9k	10k	10,980
Root	86.4	89.5	90.8	91.1	92.2	91.5	91.4	91.3	91.8	91.6	91.5
Exact	61.9	67.6	69.0	69.1	69.6	69.0	69.6	70.0	69.4	69.1	69.1
POS	84.7	86.8	87.2	87.3	86.3	86.0	85.3	86.1	85.1	85.5	85.2
Score	0.742	0.785	0.799	0.801	0.809	0.802	0.805	0.806	0.806	0.803	0.803

Validation accuracy (in %, 1,736 words) rises quickly to step 3,000 and then moves by less than one point. The best score comes at step 5,000 (0.8088); the final step scores 0.8033, so the last half of training mainly fits the training set further (loss about 0.01) without improving held-out words. Run B matched run A at step 3,000 on root accuracy (90.8% both) but was lower on exact analysis (67.3% vs 69.0%); B was cancelled, so no conclusion about pretraining is drawn.

4. Validation results of the selected checkpoint

Validation words come from two sources: **gold paradigm** cells (hand-checked tables) and **book/unified** rows (forms transcribed from printed sources). Root accuracy counts the predicted root only; exact analysis requires the root and every feature to be right.

Source / class	Words	Root	Exact analysis	Part of speech
gold paradigms				
adjective	503	92.6%	74.0%	85.3%
noun	440	88.6%	76.1%	84.1%
verb	221	95.5%	77.8%	99.1%
book / unified rows				
verb	270	91.9%	76.3%	97.4%
adjective	193	95.3%	46.1%	90.2%
noun	61	98.4%	37.7%	45.9%
infinitive	20	95.0%	15.0%	5.0%
particle	9	88.9%	66.7%	88.9%
pronoun	7	57.1%	0.0%	28.6%
demonstrative	4	75.0%	0.0%	25.0%
preposition	4	100.0%	25.0%	25.0%
numeral	3	33.3%	33.3%	66.7%
nominal	1	100.0%	100.0%	100.0%
all validation words	1,736	**92.1%**	**69.6%**	**86.3%**

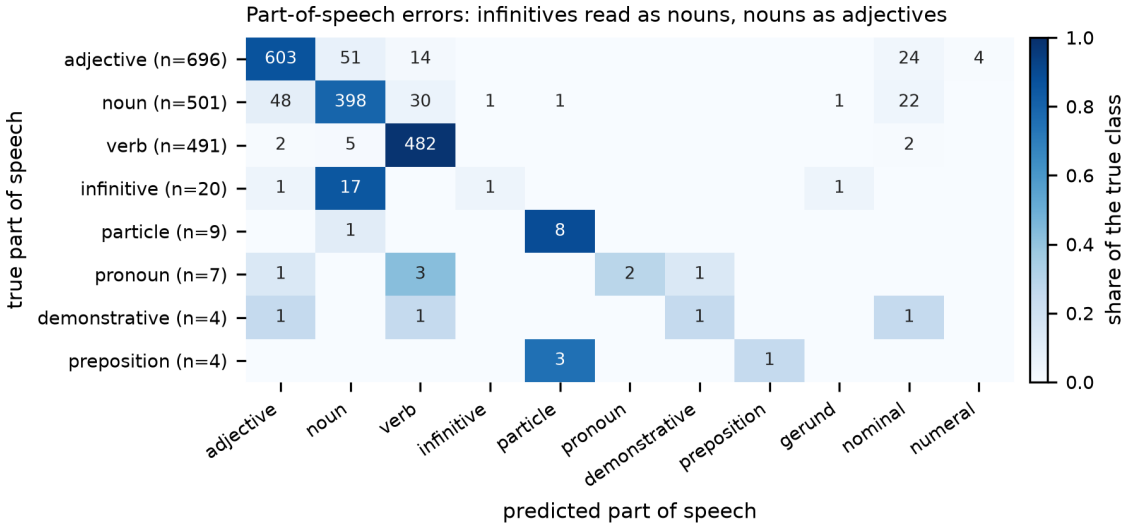


Figure 2. Part-of-speech confusion on validation words (classes with at least 4 words; counts in cells, colour = share of the true class).

5. Held-out test results

The test split was scored once, with the selected checkpoint and the same greedy decoding and scoring rule as validation, on the dataset snapshot the model was trained on (geez-dataset 04f45a7). Its roots never occur in training (leakage audit CLEAN).

Source / class	Words	Root	Exact analysis	Part of speech
gold paradigms				
adjective	627	90.4%	73.0%	88.4%
noun	580	86.7%	72.1%	79.8%
verb	336	93.5%	77.7%	97.3%
all gold test	1,543	89.7%	73.7%	87.1%
book / unified rows				
verb	363	91.2%	73.6%	96.4%
adjective	259	90.3%	39.4%	87.6%
noun	82	87.8%	37.8%	48.8%
infinitive	28	92.9%	17.9%	0.0%
particle	9	100.0%	66.7%	77.8%
pronoun	8	87.5%	50.0%	87.5%
preposition	6	100.0%	50.0%	50.0%
nominal	5	100.0%	40.0%	40.0%
all unified test	767	90.1%	54.8%	82.9%
all test words	2,310	**89.8%**	**67.4%**	**85.7%**

Book classes with fewer than 5 test words are omitted from the class rows but counted in the totals. Re-scoring on the current dataset (2c67ac1, 2,302 words after later corrections) gives root 89.8%, exact 67.7% and part of speech 85.7%, the same within 0.3 points.

Validation vs test. Test is 2.3 points lower on root accuracy and 2.2 points lower on exact analysis than validation. Part of the gap is expected, because validation chose the checkpoint; the test numbers are the ones to report.

6. What the numbers mean

• **Roots are reliable.** 90% of test words get the right root, although no test root occurs in training. This is the property the translation memory needs, because memory slots are tied to roots.
• **Full analyses are harder.** 70% are exactly right. Gold paradigm cells are easier (74–78% by class) than book rows (15–76% for classes with at least 20 words), which are more varied and closer to running text.
• **Weak spots.** Infinitives are almost always tagged as nouns (17 of 20); nouns are tagged as adjectives in 48 of 501 cases; pronouns, prepositions and demonstratives are rare in training (under 0.2% of samples each) and are often wrong.
• **Over-fitting is mild.** Training loss keeps falling to about 0.01 while validation stays flat after about 3,000 steps; about a third of the compute would have given almost the same model.

7. Limitations

1. Test and validation words are dictionary and paradigm forms, not words in running sentences; accuracy on AGE sentences can be lower, especially for names and clitics.
2. The session-2 training-loss log (steps 3,677–7,530) was not downloaded; its evaluations are included (eval_log.jsonl).
3. Synthetic forms dominate training (85% of word pairs), and verbs make up 75% of samples, which explains the weaker results on rare classes.
4. Run B (with Ge'ez pretraining) was cancelled, so the effect of pretraining is unknown.

8. Next steps

1. Re-label the AGE corpus with this model and compare the new labels with the older analyser labels used so far in the translation experiments (coverage, agreement, root changes), then rebuild the memory label caches from it (translation run G3).
2. Use its part-of-speech output as an additional signal for translation (for example, a part-of-speech slot bank or word-order features).
3. Add training data for the weak classes (infinitives, pronouns, prepositions, demonstratives).
4. Run B on TPU and compare A and B at matched steps.